Experiment 2

Aim : Setup Git and explore commands related to Version Control System (VCS)

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(VCS) Step 1: Download from https://git-scm.com
Step 2: Configuration


C:\Users\Mahipal>git config --global user.name "Mahipalsinh"

C:\Users\Mahipal>git config --global user.email"vadhermahipalsinh641@gmail.com"

C:\Users\Mahipal>git --version
git version 2.49.0.windows.1

C:\Users\Mahipal>git init
Initialized empty Git repository in C:/Users/Mahipal/.git/

C:\Users\Mahipal>mkdir ost

C:\Users\Mahipal>mkdir ost
A subdirectory or file ost already exists.

C:\Users\Mahipal>git init
Reinitialized existing Git repository in C:/Users/Mahipal/.git/

C:\Users\Mahipal>cd ost

C:\Users\Mahipal\ost>git init
Initialized empty Git repository in C:/Users/Mahipal/ost/.git/

C:\Users\Mahipal\ost>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

C:\Users\Mahipal\ost>echo sample.txt
sample.txt

C:\Users\Mahipal\ost>echo demo.py
demo.py
C:\Users\Mahipal\ost>git status
On branch master
No commits yet

nothing to commit (create/copy files and use "git add" to track)
C:\Users\Mahipal\ost>echo
ECHO is on.

C:\Users\Mahipal\ost>dir
 Volume in drive C is WIN
 Volume Serial Number is BAE2-0B5C
 Directory of C:\Users\Mahipal\ost
```

```
29-01-2026  12:19    <DIR>          .
29-01-2026  12:18    <DIR>          ..
               0 File(s)              0 bytes
 2 Dir(s)  27,356,266,496 bytes free

C:\Users\Mahipal\ost>echo Hello>sample.txt

C:\Users\Mahipal\ost>dir
 Volume in drive C is WIN
 Volume Serial Number is BAE2-0B5C

 Directory of C:\Users\Mahipal\ost

29-01-2026  12:23    <DIR>          .
29-01-2026  12:18    <DIR>          ..
29-01-2026  12:23                 7 sample.txt
               1 File(s)              7 bytes
               2 Dir(s)  27,355,930,624 bytes free

C:\Users\Mahipal\ost>type sample.txt
Hello

C:\Users\Mahipal\ost>git commit -m "Initial Commit"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        sample.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Mahipal\ost>git config --global init.defaultBranch

C:\Users\Mahipal\ost>git commit -m "Initial Commit"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        sample.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\Mahipal\ost>git add sample.txt

C:\Users\Mahipal\ost>git checkout
fatal: You are on a branch yet to be born

C:\Users\Mahipal\ost>git checkout main
error: pathspec 'main' did not match any file(s) known to git

C:\Users\Mahipal\ost>git checkout -b dev
```

```
Switched to a new branch 'dev'

C:\Users\Mahipal\ost>echo hello from dev>>sample.txt

C:\Users\Mahipal\ost>git add .

C:\Users\Mahipal\ost>git commit -m "Update in dev"
[dev (root-commit) fd94f73] Update in dev
 1 file changed, 2 insertions(+)
 create mode 100644 sample.txt

C:\Users\Mahipal\ost>git merge dev
Already up to date.

C:\Users\Mahipal\ost>dir
 Volume in drive C is WIN
 Volume Serial Number is BAE2-0B5C

 Directory of C:\Users\Mahipal\ost

29-01-2026  12:23    <DIR>          .
29-01-2026  12:38    <DIR>          ..
29-01-2026  12:40                23 sample.txt
               1 File(s)             23 bytes
               2 Dir(s)  28,156,104,704 bytes free

C:\Users\Mahipal\ost>type sample.txt
Hello
hello from dev

C:\Users\Mahipal\ost>git remote -v
origin  https://github.com/mahipal1075/ost (fetch)
origin  https://github.com/mahipal1075/ost (push)


C:\Users\Mahipal\ost>git branch
* dev

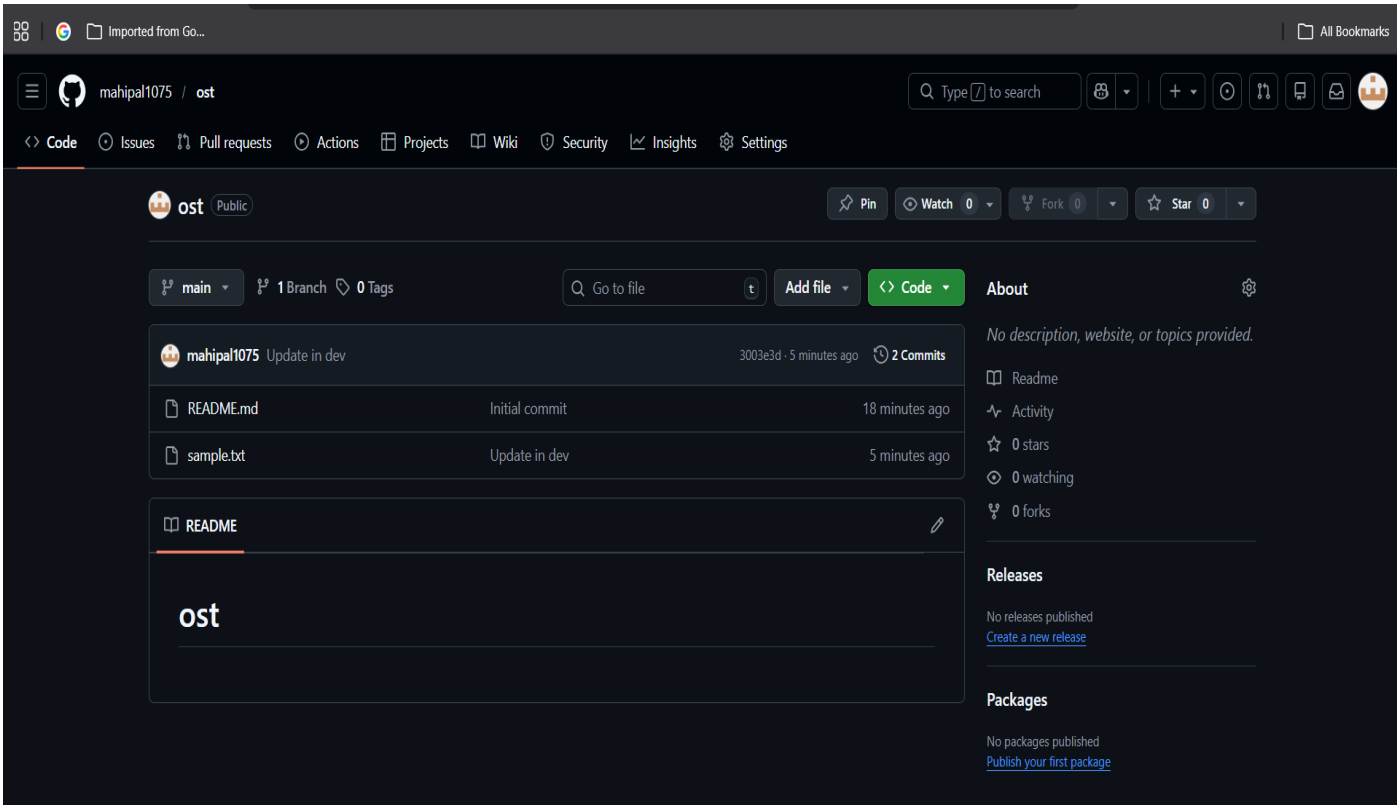
C:\Users\Mahipal\ost>git branch -M main

C:\Users\Mahipal\ost>git push origin main
To https://github.com/mahipal1075/ost
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/mahipal1075/ost'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

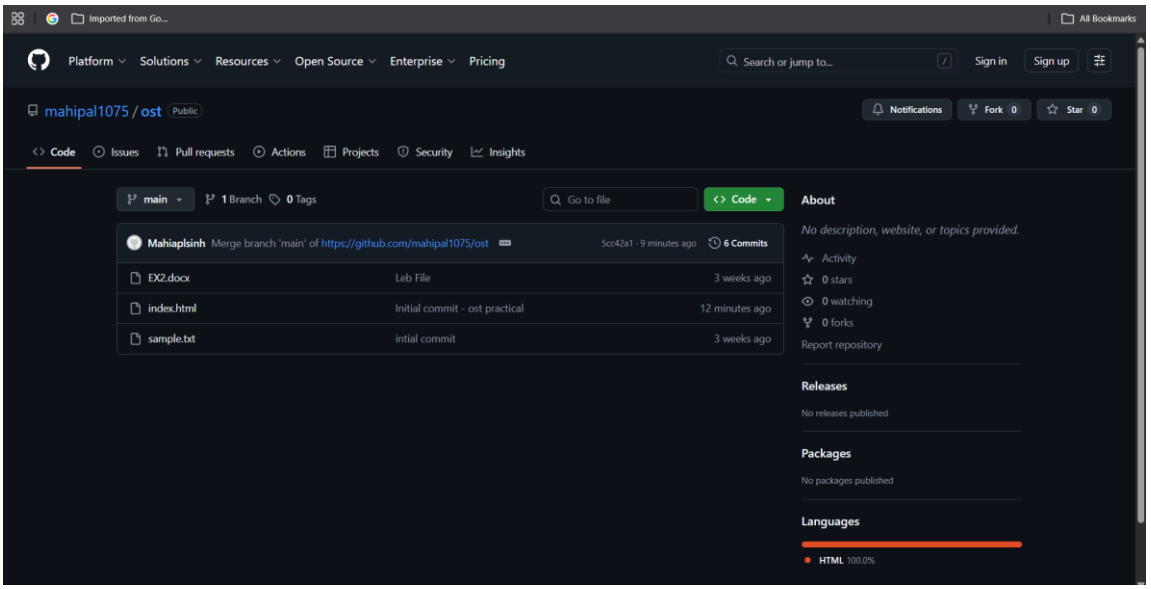

C:\Users\Mahipal\ost>git branch
* main
C:\Users\Mahipal\ost>git pull --rebase origin main
From https://github.com/mahipal1075/ost
 * branch            main       -> FETCH_HEAD
```

```
Successfully rebased and updated refs/heads/main.

C:\Users\Mahipal\ost>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 325 bytes | 325.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mahipal1075/ost
   dddd0de..3003e3d  main -> main
branch 'main' set up to track 'origin/main'.
```



Prectical 3

Aim : Create a GitHub/GitLab repository and upload sample code

```
C:\Users\Mahipal>git config --global user.name "Mahiaplsinh"

C:\Users\Mahipal>git config --global user.email "mahipalsing123@gmai.com"

C:\Users\Mahipal>mkdir ost_practical

C:\Users\Mahipal>cd ost_practical

C:\Users\Mahipal\ost_practical>notepad index.html

C:\Users\Mahipal\ost_practical>git init
Initialized empty Git repository in C:/Users/Mahipal/ost_practical/.git/

C:\Users\Mahipal\ost_practical>git add .

C:\Users\Mahipal\ost_practical>git commit -m "Initial commit - ost practical"
[master (root-commit) 4e97cba] Initial commit - ost practical
 1 file changed, 98 insertions(+)
 create mode 100644 index.html

C:\Users\Mahipal\ost_practical>git remote add origin https://github.com/mahipal1075/ost

C:\Users\Mahipal\ost_practical>git branch -M main

C:\Users\Mahipal\ost_practical>git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.44 KiB | 735.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mahipal1075/ost
   c003493..5cc42a1  main -> main
```